



High-performance thin-film lithium niobate electro-optic modulator on a thick silica buffering layer

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Abstract: High-speed thin-film lithium niobate (TFLN) electro-optic modulators are essential for next-generation photonic integrated circuits. We demonstrate a high-performance TFLN Mach–Zehnder modulator on a thick silica buried layer using periodic capacitively loaded traveling-wave electrodes. Our design simplifies fabrication compared to undercut silicon or quartz substrates while enhancing modulation efficiency. The fabricated device achieves a modulation efficiency of 1.25 V·cm in the C-band with only 1.3 dB roll-off at 67 GHz, successfully overcoming the voltage-bandwidth trade-off. This work provides a streamlined, scalable approach for high-performance TFLN modulators, supporting advanced optical communications and signal processing.

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1. Introduction

Over the past decades, the rapid advancement of the information age, driven by cutting-edge technologies such as 5G/6G networks, artificial intelligence (AI), ChatGPT, and cloud computing, has triggered an unprecedented surge in global data traffic [1,2]. This exponential growth has pushed the limits of data processing speed, operational bandwidth, and power efficiency in signal processing devices [3], and posed significant challenges to the conventional electronic integrated circuits [4]. To address these challenges, the ultra-high-speed photonic integrated circuits (PICs) have emerged as a next-generation solution, enabling efficient EO signal conversion and processing and ‘instantaneous’ manipulation of massive data streams [5,6]. EO modulators, which are the paramount component of such PICs, play a critical role by bridging microwave signals ($\sim 10^9$ Hz) and optical carrier signals ($\sim 10^{14}$ Hz) across different frequency domains. These modulators are fundamental to high-performance data transmission, interconnection, processing, and computation [7,8]. For PICs to achieve widespread adoption, the development of ultra-high-performance EO modulators is essential. Key requirements include low driving voltage, broad bandwidth, minimal power consumption, negligible insertion loss, high extinction ratio (ER), and compatibility with large-scale manufacturing.

Over the years, EO modulators have been extensively studied across various photonic platforms, including silicon [7,9], lithium niobate (LN) [10–16], lithium tantalate (LT) [5], hybrid silicon nitride (Si_3N_4) [17–22], indium phosphide (InP) [23,24], polymers [25–27], and plasmonics [28–30]. Each platform offers distinct advantages, such as well-established fabrication processes (Si), exceptionally low propagation losses (hybrid Si_3N_4), capability for dense integration (Si, hybrid Si_3N_4 , InP), high modulation efficiency (Si, InP, polymers, plasmonics), large bandwidth (LN, LT, polymers), and high linearity (LN, LT) [31]. Nevertheless, developing high-performance EO modulators that simultaneously meet all critical performance metrics remains a significant challenge.

Offering unparalleled performance, thin-film lithium niobate (TFLN) has recently emerged as a leading platform for next-generation photonic integrated circuits (PICs). The improvement arises from strong optical confinement of TFLN nanowaveguides while leveraging LN's exceptional properties, including a broad transparent range (0.35–5 μm), large EO coefficient (~ 30 pm/V), and strong nonlinearity (~ 27 pm/V) [31,32]. The synergy significantly enhances EO interaction, enabling compact EO modulators with CMOS-compatible driving voltages [13,33–35]. Previous demonstrations of TFLN EO modulators using coplanar waveguide (CPW) electrodes have achieved remarkable performance, with modulation efficiency around 2.5 V·cm and bandwidths from 40 GHz to 67 GHz [10–12]. These advances have enabled breakthroughs in large-capacity optical telecommunications [8,12], EO frequency combs [36,37], and synthetic dimensions research [38–40]. However, the fundamental voltage-bandwidth limit remains constrained by the need to simultaneously optimize EO overlap, group velocity matching, and microwave propagation loss. Innovative micro-structured electrodes have recently overcome this limitation, enabling TFLN modulators to achieve both high efficiency (~ 2.1 V·cm) and broad 3-dB modulation bandwidths (> 100 GHz) [15,19,41–46]. These designs typically employ either partially etched silicon or lithium-niobate-on-quartz (LNOQ) substrates for optimal group velocity matching. However, partially etched silicon introduces fabrication complexity that hinders mass production, while LNOQ approaches require thicker upper cladding for microwave index tuning, which reduces modulation efficiency and creates fabrication challenges including conductivity issues in electron beam lithography (EBL) and the need for specialized end-face treatments.

Here, we present a high-performance TFLN EO MZM featuring periodic capacitively loaded traveling-wave (CLTW) electrodes and a 12- μm -thick silica buried layer on silicon substrate. Notably, the device exhibits exceptional performance with 1.25 V·cm modulation efficiency in the telecom C-band, achieved through optimized design of narrow T-rails gap and reduced upper cladding thickness. Furthermore, we undertake investigate the relationship between the microwave index and electrode configuration, and propose an innovative strategy to achieve group velocity matching. These findings not only advance current modulator design but also establish critical design guidelines for future high-performance EO devices. The proposed methodology provides both fundamental insights and practical engineering references for next-generation modulator development. Our device with a 7-mm-long interaction region demonstrates excellent high-frequency performance with only 1.3 dB roll-off in EO response at 67 GHz. Our approach successfully overcomes the traditional voltage-bandwidth trade-off in TFLN modulators while maintaining fabrication simplicity, which is a critical advantage for practical implementation. The proposed TFLN EO modulator holds great promise for applications in the next-generation PICs.

2. Device design and fabrication

The proposed TFLN EO MZM with CLTW electrodes in the push-pull configuration is schematically illustrated in Fig. 1(a), showing its implementation on a thick silica-buffered silicon substrate. Figure 1(b) details the cross-sectional structure of the modulation region, fabricated on a commercial x-cut TFLN wafer (NANOLN) consisting of a 360-nm-thick LN layer bonded

to a 12- μm -thick silica buffer layer atop a standard 500- μm silicon substrate, see also inset of Fig. 1(b).

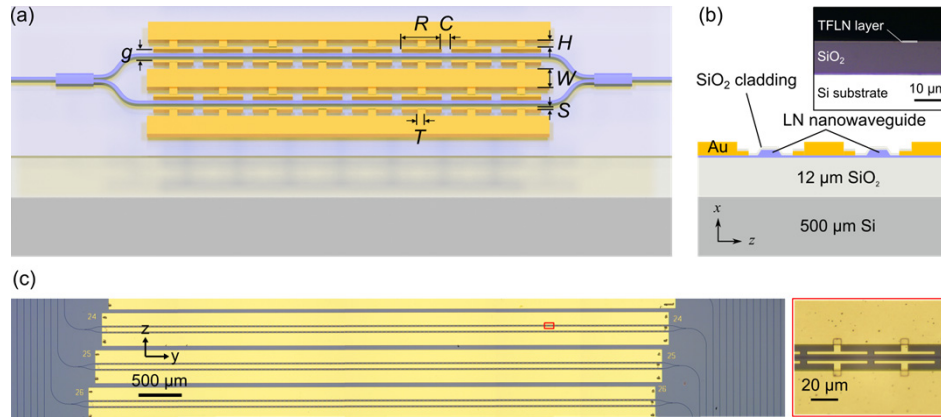


Fig. 1. TFLN MZM based on a thick silica buried layer. (a) Schematic of the TFLN MZM. In our optimal design, W , g , H , S , T , R , and C are 50, 5, 5.2, 2, 5, 45, and 5 μm , respectively. (b) Cross section of the modulator (not to scale). Inset: Optical micrograph of the cross section of the TFLN on 12- μm -thick SiO_2 buffering layer. (c) Optical microscopy image of the fabricated chip. Inset: Image of the T-rails micro-structured electrodes.

The fabrication begins with defining the Mach-Zehnder interferometer pattern using electron beam lithography (EBL), followed by inductively coupled plasma (ICP) etching of the TFLN layer to create ridge nanowaveguides with 260 nm etching depth, 60° sidewall angle, and 800 nm topwidth. For the CLTW electrodes, we employ a two-step process: Firstly, the T-rails electrodes (600-nm Au/10-nm Ti) are patterned using high-resolution EBL with polymethylmethacrylate (PMMA) resist and electron beam evaporation (EBE); then, the main electrodes (1.1- μm Au/20-nm Ti) are fabricated through UV lithography and a second evaporation step (The CLTW electrode patterning can also be achieved using one-step DUV lithography). This specific two-step approach is motivated by several considerations: (1) The high-resolution patterning required for the narrow-gap T-rails is optimally achieved with EBL, while the larger main electrodes can be efficiently fabricated using UV lithography. (2) The chosen T-rail thickness (600 nm) represents a balance between achieving low microwave resistance and maintaining reliable lift-off process fidelity. (3) The thicker main electrodes (1.1 μm) are designed to significantly reduce microwave conductor loss, a thickness readily achievable via the subsequent UV lithography and evaporation step. This strategy allows for independent optimization of lithography method and metal thickness for different electrode components, ensuring both high performance and fabrication robustness. To achieve group velocity matching, we deposit a 100-nm silica upper cladding layer by plasma-enhanced chemical vapor deposition (PECVD) at 400°C and 200 mTorr using SiH_4 and O_2 precursors. The 100-nm thickness of the silica upper cladding is selected as the optimal value from finite-element simulations among multiple thicknesses (i.e., 100, 300, 500, and 700 nm), which shows the best group-velocity matching between the microwave and optical signals. The performance is found to be sensitive to this parameter, with thicker claddings resulting in degradation of the high-frequency response due to velocity mismatch. Ellipsometry and profilometry measurements confirm excellent layer uniformity with $<5\%$ thickness variation across the structure. Subsequent ICP-RIE etching with CHF_3/Ar at 20°C (50 nm/min rate) exposes the main electrodes while maintaining good RF probe contact. The chip end facets are precision-polished for optimal light coupling. The fabrication processes are carried out at the Center for Advanced Electronic Materials and Devices (AEMD), Shanghai Jiao Tong University.

Figure 1(c) presents an optical micrograph of the completed device. In operation, TE0 mode input light couples via lensed fiber and splits equally by a 50:50 multimode interferometer (MMI) into both MZI arms. The 7-mm-long push-pull phase shifters enable co-propagation with microwave signals in the transmission line electrodes, with the phase-modulated light recombined by an identical output MMI to generate intensity-modulated output. This carefully engineered structure combines optimized waveguide geometry, precisely controlled electrode configuration, and tailored dielectric layers to achieve superior EO performance.

Although TFLN-based modulators benefit from easier group velocity matching compared to bulk counterpart due to the low relative permittivity (~ 11.9) of silicon substrates, they suffer from significantly increased microwave losses. These losses originate from two primary sources, metal loss caused by finite conductor resistivity, and dielectric loss from the buried layer and substrate absorption. A critical trade-off emerges in device design. While reducing the electrode gap below $5\text{ }\mu\text{m}$ enhances modulation efficiency (half-wave voltage length product $V_\pi \cdot L$), it simultaneously exacerbates metal losses due to current crowding effects. To address this fundamental challenge, we implement a periodic CLTW electrode design, as shown in Fig. 1(c). This configuration features a CPW structure with micro-patterned T-rails that extend from the main electrodes into the gap region. The T-rails serve a dual purpose: they maintain high modulation efficiency while effectively preventing the current from crowding on the electrode edges near the gap region. This strategic optimization substantially reduces microwave losses without sacrificing the crucial electrode gap-dependent modulation efficiency.

Notably, the CLTW electrode configuration demonstrates a significantly increased microwave index compared to conventional CPW designs. This improvement stems from the T-rails' ability to increase capacitance per unit length while maintaining relatively constant inductance, thereby boosting the microwave refractive index through the slow-wave effect. While previous researches have employed low-permittivity quartz substrates (~ 4) or partially etched silicon substrates to address the high microwave index and achieve velocity matching [41,42,44,45], these solutions introduce substantial practical limitations. For instance, quartz-based implementations face fundamental challenges as their inherently low microwave index often falls below the optical group index, requiring excessively thick silica cladding layers for proper velocity matching. Moreover, LNOQ platforms suffer from higher optical propagation losses (attributed to lower substrate thermal conductivity induced poorer etching quality), reduced fabrication yield, which is detrimental to large-scale integration or heterogeneous integration with silicon photonics. Similarly, partially etched silicon substrates present formidable fabrication hurdles, demanding nanometer-scale precision in both etching depth and interlayer alignment throughout the manufacturing process.

In contrast, our approach utilizes TFLN with a $12\text{-}\mu\text{m}$ -thick silica buffer layer, which can effectively reduce the microwave refractive index to better match that of guided optical modes while using low-loss CLTW electrodes. This configuration achieves an optimal balance by maintaining the microwave refractive index at an appropriately elevated level, enabling perfect group velocity matching with only a thin upper cladding layer. Compared to LNOQ platforms and undercut-etched silicon substrates, our solution offers significantly simplified fabrication while maintaining performance. Notably, a thin silica cladding layer enables an improved modulation efficiency. This can be attributed to the fact that the electric field at the LN-silica interface is smaller than the air-silica interface, which leads to a stronger EO overlap.

To verify the velocity matching characteristics of our CLTW electrode design on TFLN with $12\text{-}\mu\text{m}$ -thick silica buffer layers, we conduct simulations using finite element methods (Ansys HFSS) to assess the microwave index of various electrode configurations: conventional CPW electrodes on TFLN with $4.7\text{-}\mu\text{m}$ silica buffer, CPW on TFLN with $12\text{-}\mu\text{m}$ silica buffer, CLTW electrodes on TFLN with $12\text{-}\mu\text{m}$ silica buffer, and CLTW electrodes on LNOQ platforms. All models incorporate a 100-nm silica upper cladding layer, with electrode parameters carefully

extracted from relevant Refs. [15,17]. As shown in Fig. 2(a), our simulations reveal two key findings: First, increasing the silica buffer thickness from 4.7 μm to 12 μm effectively compensates for the microwave index difference between CLTW and CPW configurations. Second, the TFLN platform with 12- μm buffer demonstrates superior velocity matching capability compared to LNOQ implementations. This 12- μm thickness significantly reduces dielectric losses at low frequencies compared to thinner ($\leq 4.7 \mu\text{m}$) buffers, thereby optimizing the response of the electrode. On the other hand, the CLTW electrode exhibits a substantial reduction in ohmic losses, leading to a notable optimization for the high-frequency response of the electrode. Compared to LNOQ, our thin silica upper cladding suffices to achieve velocity matching condition, ensuring a high modulation efficiency.

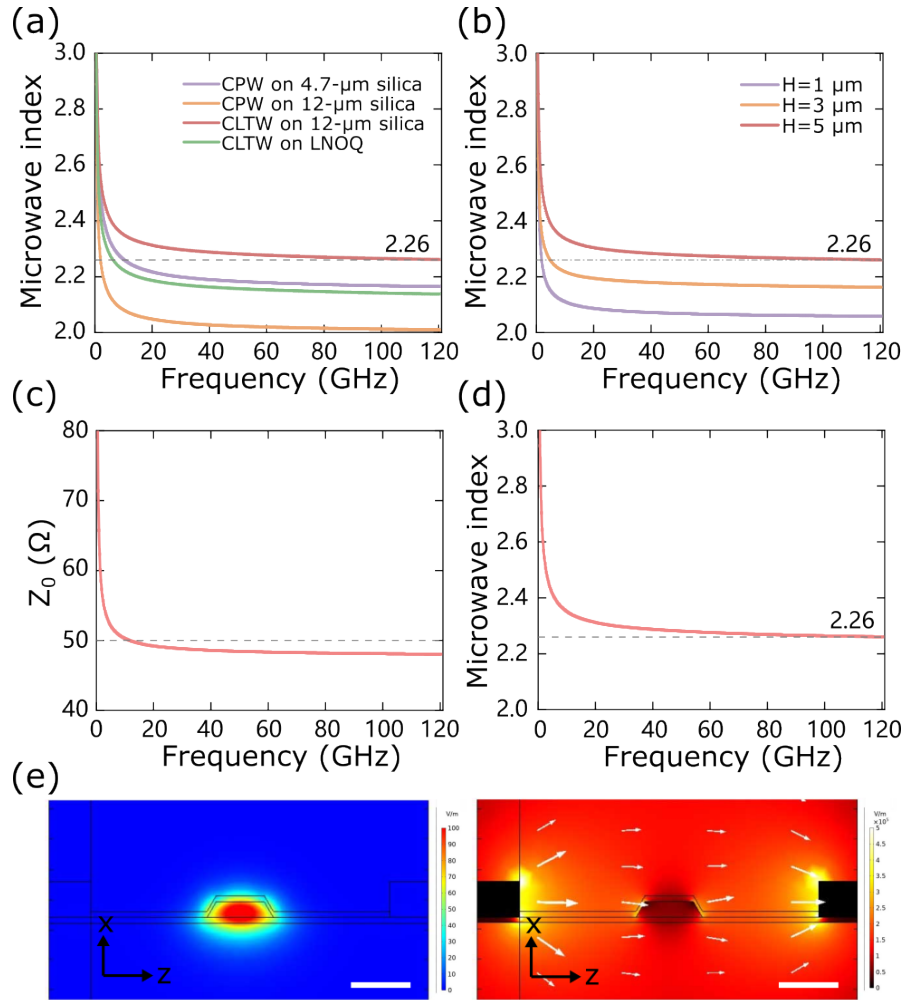


Fig. 2. (a) Simulated microwave indices under different electrode configurations on different buried layer or substrate. (b) Simulated microwave indices for different H parameters. (c) Calculated characteristic impedance Z_0 . (d) Microwave index of the CLTW electrode. (e) Simulated optical and electrical field distribution. Scaler bar: 1 μm .

The microwave index exhibits a direct positive correlation with capacitance per unit length, providing an effective approach for precise index tuning. Through systematic adjustment of

T-rail geometric parameters, the capacitance of the CLTW electrode can be adjusted to achieve a precise microwave index necessary for perfect velocity matching when the parameter H is equal to 1, 3 and 5 μm , as illustrated in Fig. 2(b). Through comprehensive simulations, we optimize the CLTW electrode geometry with parameters $(W, g, H, S, T, R, C) = (50, 5, 5.2, 2, 5, 45, 5) \mu\text{m}$. The resulting characteristics, as shown in Figs. 2(c) and 2(d), demonstrates tolerant impedance matching and perfect velocity matching. Figure 2(e) presents the simulated optical mode profile of the transverse electric mode and the electrostatic field of the microwave electrodes. These design optimizations yield a calculated group index of light of 2.26 and modulation efficiency of 1.20 V·cm at 1550 nm [31].

3. Measurement and discussion

Figure 3(a) shows the experimental setup and the fabricated modulator devices. Optical characterization reveals an on-chip insertion loss of approximately 1.0 dB for the MZM, which includes contributions from both the input/output multimode interference couplers (MMIs) and waveguide propagation losses. Using the Fabry-Perot (F-P) interferometric method [47], we measured the waveguide propagation loss to be approximately 0.3 dB/cm at 1550 nm, based on a reference waveguide on the same chip. This low propagation loss remains comparable to waveguides without metal electrodes, demonstrating the minimal optical degradation introduced by our electrode design.

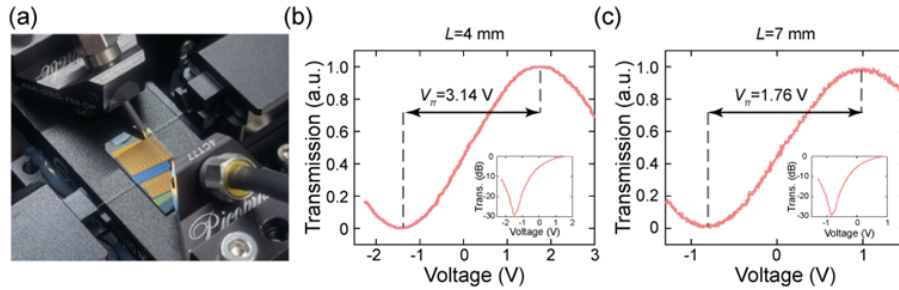


Fig. 3. (a) The TFLN MZM chip sample under test. Normalized optical transmission of the (b) 4- and (c) 7-mm-long modulators as a function of the applied voltage, showing a half-wave voltage of 3.14 V and 1.76 V, respectively. The insets show the measured transmissions in a logarithmic scale showing the ERs of the corresponding devices.

The static characterization of our modulators at the wavelength of 1550 nm reveals outstanding performance metrics. For devices with 4-mm and 7-mm modulators, as depicted in Figs. 3(b) and 3(c), we measure the V_π values of 3.14 V and 1.76 V respectively, corresponding to the $V_\pi \cdot L$ product of $\sim 1.25 \text{ V}\cdot\text{cm}$, which aligns well with theoretical prediction. This represents an improvement in modulation efficiency compared to other CLTW electrode-based devices listed in Table 1. Further voltage reduction below 1 V remains achievable through either electrode gap reduction or modulation length extension. Additionally, the insets in Figs. 3(b) and 3(c) illustrate the device transmissions on a logarithmic scale, revealing extinction ratios of approximately 30 dB, as determined through a static voltage sweep measurement. This relatively high performance stems from our optimized fabrication approach through precise electrode alignment enabled by EBL and improved write-field stitching compared to LNOQ-based implementations.

Subsequently, we characterize the electrical-electrical (EE) S parameters of the CLTW electrode, as depicted in Fig. 4(a). The EE S_{21} measurements demonstrate excellent broadband response, with S_{21} showing only -6.4 dB roll-off at 67 GHz for the 7-mm modulator. The EE reflection coefficient S_{11} remains below -21 dB across the entire measured spectrum, indicating minimal

Table 1. Comparison of performance metrics for TFLN MZMs

Electrode (Substrate)	$V_{\pi} \cdot L$ (V·cm)	Length (mm)	EO roll-off	ER (dB)	Ref.
CPW (Si)	2.8	20	~3 dB at 45 GHz	30	[10]
CPW (Si)	1.29	3	<3 dB at 40 GHz	19	[16]
CPW (Si)	1.02	5	3 dB at 108 GHz	/	[13]
CPW (hybrid)	2.55	5	~3 dB at 70 GHz	/	[17]
Homologous CLTW (hybrid)	3.1	5	~3 dB at 110 GHz	28	[18]
CLTW (quartz)	2.3	10	1.8 dB at 50 GHz	20	[15]
CLTW (undercut etched Si)	2.2	10	~1.4 dB at 67 GHz	>20	[41]
CLTW (undercut etched Si)	6.2	20	3 dB at 50 GHz	24	[42]
CLTW (undercut etched Si)	2.125	12.5	~2 dB at 60 GHz	25	[48]
This work	1.25	7	1.3 dB at 67 GHz	~30	-

signal reflection. The extracted characteristic impedance ($Z_0 \approx 46 \Omega$) shows slight deviation from the ideal impedance matching. However, the extracted characteristic impedance encompasses contributions from all output ports, microwave cables, and RF probes, each featuring a 50- Ω characteristic impedance, suggesting that the intrinsic Z_0 of the CLTW electrodes might be less than 46 Ω . This suggests the intrinsic CLTW electrode impedance may actually be lower than 46 Ω , representing even better matching to standard 50 Ω designs. Figure 4(c) demonstrates excellent velocity matching in the high-frequency regime, with the microwave index n_m converging to the group index of the light (~ 2.26).

To quantitatively demonstrate the microwave loss reduction achieved by our CLTW electrode design, we performed a direct comparison with conventional CPW electrodes fabricated under identical conditions with matching gap dimensions. As shown in Fig. 4(d), the measured microwave losses are 0.74 dBcm-1GHz-1/2 for the CLTW electrode versus 1.05 dBcm-1GHz-1/2 for the CPW electrode, representing a significant 30% reduction in loss through the incorporation of T-rails. The microwave losses are substantially lower than the average values reported in the literature, confirming our CLTW-based modulator as a leading candidate for ultra-high-performance photonic applications.

The small signal EO response of the device is characterized using a vector network analyzer (Agilent, N4373E). The RF signal from the VNA is applied to the electrode via a high-speed RF probe, while the other end of the electrode is connected to a 50- Ω termination through another probe. As shown in Fig. 5(a), the 7-mm MZM device exhibits exceptional broadband performance with a measured 3-dB EO bandwidth exceeding 67 GHz (limited by the VNA's frequency range). Based on the flat frequency response up to 67 GHz and the optimized velocity matching, the device is expected to maintain low roll-off beyond 110 GHz, though direct experimental validation would be required to confirm the exact 3-dB bandwidth. The measured EO S_{21} response reveals a rapid 1-dB roll-off at low frequency range below 2 GHz, which stems from impedance mismatch effects as evidenced by resonance ripples in both S_{21} and S_{11} responses and deviations in the extracted characteristic impedance [31]. While further optimization toward perfect 50 Ω matching could potentially reduce this low-frequency ripple, this represents a valuable direction for future design iterations and does not detract from the core high-frequency performance of the device. The EO response remains remarkably flat from 2 GHz to 67 GHz with only 1.3 dB roll-off at the maximum measured frequency, demonstrating perfect group velocity matching achieved through our optimized CLTW electrode design. The sustained flatness of the EO response at high frequencies, combined with the extrapolated bandwidth exceeding 120 GHz, confirms the device's outstanding potential for ultra-broadband operation in next-generation communication systems.

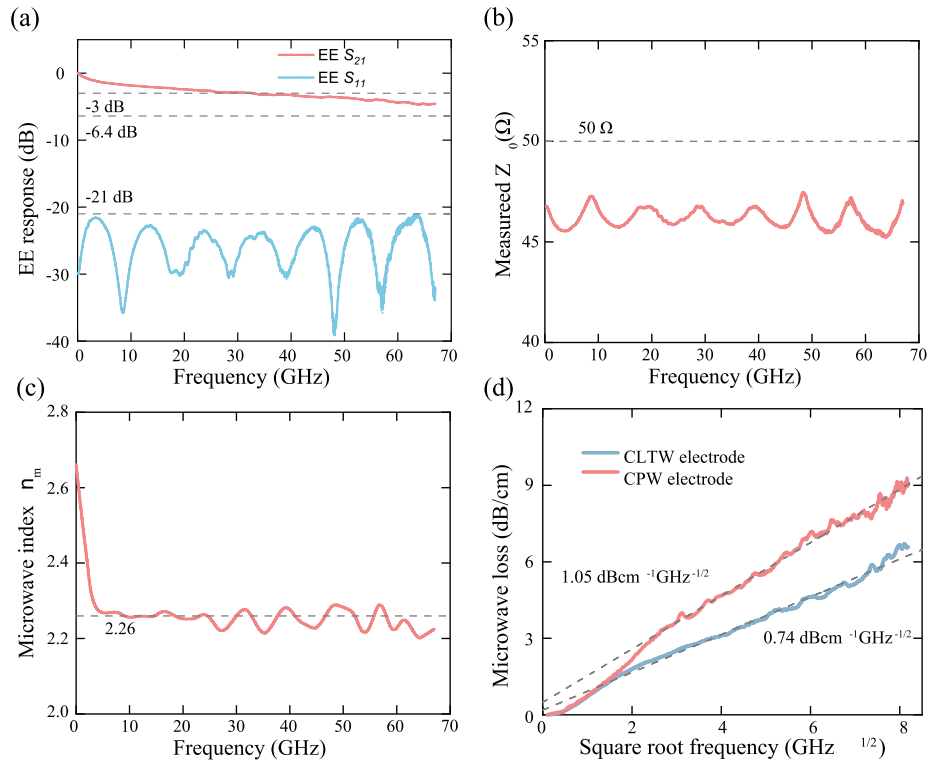


Fig. 4. (a) Measured EE coefficients of S_{21} and S_{11} for the 7-mm MZM device, indicating a 3-dB bandwidth of approximately 35 GHz and a 6-dB bandwidth exceeding 67 GHz. Extracted (b) characteristic impedance Z_0 and (c) microwave index n_m for the CLTW electrode. (d) Measured microwave losses for both CLTW electrode and CPW electrode.

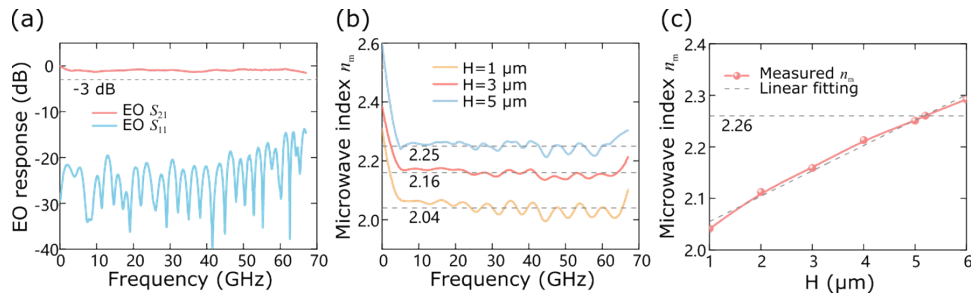


Fig. 5. (a) Measured EO responses S_{21} and electrical reflection S_{11} of the fabricated 7-mm-long MZM device, showing a high 3-dB bandwidth significantly surpassing 67 GHz. (b) Calculated microwave index values for different H parameters. (c) Correlation between microwave index and H parameters.

Furthermore, we investigate a precise method to finely tune the microwave index to achieve group velocity matching based on the CLTW electrode. As illustrated in Fig. 1(a), the H parameter serves as a critical design variable that directly influences the electrode capacitance and consequently the microwave refractive index. Through systematic analysis, we extract and compare the microwave index for H values of 1, 3, and 5 μm , as shown in Fig. 5(b), revealing a well-defined linear relationship at high frequencies with a slope of $0.05 \mu\text{m}^{-1}$ in Fig. 5(c). This linear dependence enables accurate tuning of the microwave index by controlling the T-rail geometry, providing a reliable pathway to achieve optimal group velocity matching.

The high-speed data transmission capabilities of our 7-mm MZM device are rigorously evaluated, as depicted in Fig. 6(a). A 65-GHz bandwidth arbitrary waveform generator (Keysight M8199A) generated the electrical data signal, which is then amplified by a RF amplifier. The amplified signal, combined with a DC bias via a Bias-T, is used to drive the modulator, and the output signal is amplified by an erbium-doped fiber amplifier (EDFA) and refined by a band-pass filter. The output light is then detected by a high-speed photodetector and the eye diagrams is recorded by an oscilloscope (Keysight N1000A) with a bandwidth of 65 GHz. For OOK modulation, clear eye openings are obtained at both 80 Gb/s and 112 Gb/s with dynamic ERs of 7.14 dB and 4.42 dB respectively, as shown in Figs. 6(b) and 6(c). Additionally, PAM-4 modulation is implemented at 40 Gbaud (80 Gb/s) and 50 Gbaud (100 Gb/s), with the results presented in Figs. 6(d) and 6(e). The dynamic ERs reach 5.05 and 5.09 dB, respectively. Notably, the highest PAM4 symbol rate presented is constrained by the analog bandwidth of the test setup, including the arbitrary waveform generator and oscilloscope. Nevertheless, the successful 112 Gbaud NRZ and 50 Gbaud Pam-4 transmission indicates its capability to support higher PAM4 symbol rates with appropriate driving and detection systems. The back-to-back (B2B) bit error rate (BER) performance is systematically characterized for both 64 Gb/s OOK and 80 Gb/s PAM-4 signal transmissions, as shown in Fig. 6(f). The measured BERs consistently remain below the KP-4 forward error correction (FEC) threshold of 2.4×10^{-4} for both modulation formats. Particularly for the 80 Gb/s PAM-4 transmission, the BER further decreases below the hard decision forward error coding (HD-FEC) threshold of 3.8×10^{-3} when the received optical power surpasses -17 dBm. The BER exhibits a linear decrease with increasing received optical power before pre-amplification, and no error floor is observed within the tested optical power range. These results unambiguously demonstrate the superior linearity and transmission quality of our modulator for high baud-rate optical communications. It is pertinent to emphasize that the current performance limitations are primarily attributed to external electronic components, including the digital-to-analog converter, RF probe, and driver noise, rather than distortions inherent to the modulator itself. This suggests significant potential for further performance enhancement through optimization of the driving electronics.

Finally, we summarize the performance metrics of recent research on TFLN MZMs in Table 1, highlighting the advantages of our approach. The demonstrated device achieves superior modulation efficiency (lower $V_\pi \cdot L$) compared to LNOQ-based and undercut-etched substrate designs, primarily enabled by our optimized thinner silica buffer layer architecture. Crucially, the velocity matching is obtained in our proposed modulator based on a 12- μm -thick silica buried layer, which is favored in volume fabrication. From a manufacturing perspective, the design provides simplified fabrication flow with higher process yield, reduced production costs through elimination of complex substrate etching steps, and maintained compatibility with standard lithographic techniques. These practical advantages, combined with the measured high-frequency performance and modulation efficiency, position our modulator as an optimal solution that successfully balances all critical performance metrics for next-generation photonic integrated circuits.

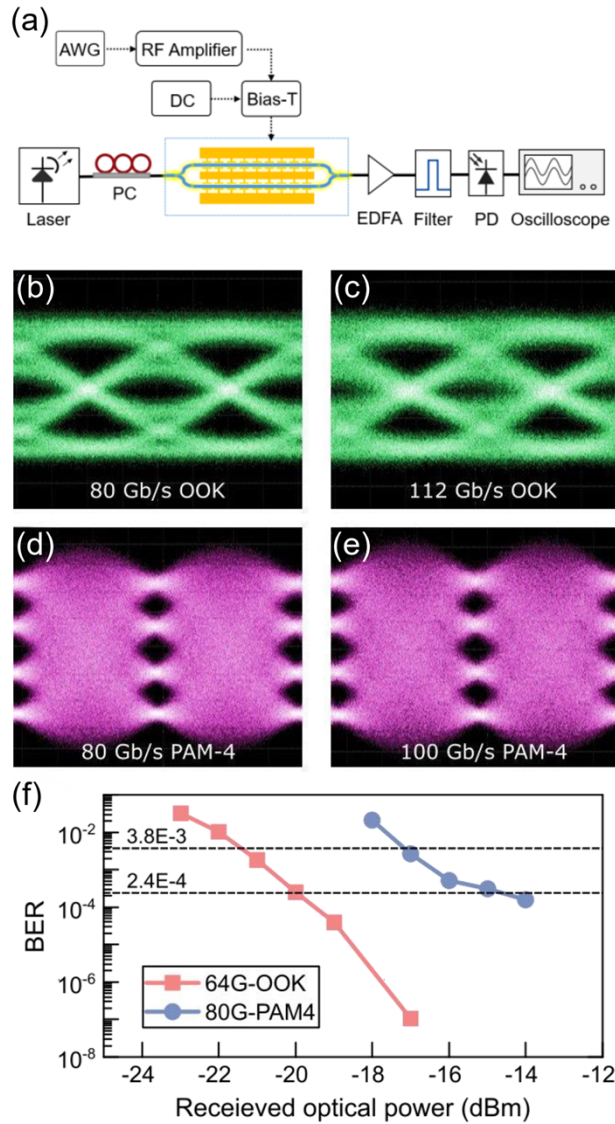


Fig. 6. High-speed data transmission of the fabricated 7-mm MZM device. (a) Schematic of the experimental setup. Measured optical eye diagrams for the OOK format at data rates of (b) 80 Gb/s and (c) 112 Gb/s. Measured optical eye diagrams for the PAM-4 format at data rates of (d) 40 Gbaud (80 Gb/s) and (e) 50 Gbaud (100 Gb/s). (f) The BER versus the received optical power for both 64 Gb/s OOK and 80 Gb/s PAM-4 signal transmissions.

4. Conclusion

This work presents a significant advancement in TFLN modulator technology through the demonstration of CLTW electrode-based MZMs on a thick silica buffer platform. Our experimental results showcase exceptional device performance, including an ultra-flat EO frequency response with merely 1.3 dB roll-off at 67 GHz and a record-high modulation efficiency of 1.25 V·cm in the telecom C-band. Moreover, we have proposed and demonstrated an efficient method to manage the microwave index to achieve perfect group velocity matching. Combined with the

significantly reduced microwave loss afforded by the CLTW electrode, this represents the first demonstration of an ultra-high-performance TFLN modulator implemented on a thick-silica platform. The developed thick-silica TFLN platform, coupled with our microwave index management methodology, establishes a new paradigm for high-performance EO devices. We anticipate these advancements will serve as fundamental building blocks for future research and development of TFLN-based PICs, accelerating their adoption in next-generation optical communication systems, microwave photonics, and quantum technologies.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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